

Linux Enumeration

Linux Enumeration Before Exploitation (External Footprinting & Scanning)

Target: Remote Linux host

Goal: Find entry points, exposed services, credentials, files, and potential vulnerabilities

0. Discovery & Scanning

◆ Basic Host Discovery

```
ping -c 1 10.10.11.99
```

◆ Port & Service Discovery

```
nmap -p- -T4 10.10.11.99  
nmap -sC -sV -T4 -p- --min-rate 5000 10.10.11.99
```

Use full TCP scan first, then targeted:

```
nmap -sC -sV -p 21,22,80,111,139,443,445,2049 10.10.11.99
```

1. Open Services Enumeration (Pre-Auth)

1.1 SSH (Port 22)

- Try default or weak credentials (`root:toor` , `admin:admin`)
- Brute-force with `hydra` :

```
hydra -l root -P rockyou.txt ssh://10.10.11.99
```

- Check for SSH banner (`nmap -sV --script ssh2-enum-algos`)

1.2 HTTP/HTTPS (Ports 80, 443)

Recon Tools

```
whatweb http://10.10.11.99
nikto -h http://10.10.11.99
nmap --script http-enum -p 80 10.10.11.99
```

Content Discovery

```
ffuf -u http://10.10.11.99/FUZZ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -c
```

Check:

- `/robots.txt` , `/backup/` , `/admin/` , `/uploads/`
- Page source, JavaScript, hidden inputs

If login page is found:

- Capture POST request & bruteforce with `hydra` or `wfuzz`

- Check for SQLi, LFI, SSTI

1.3 FTP (Port 21)

```
nmap -p 21 --script ftp-anon,ftp-bounce,ftp-syst 10.10.11.99
```

```
ftp 10.10.11.99  
# Try anonymous login
```

- Download everything (`get *`)
- Look for config files, `.bak` , `.conf` , `db.sqlite` , etc.

1.4 SMB (Ports 139, 445)

```
smbclient -L //10.10.11.99 -N  
smbmap -H 10.10.11.99  
enum4linux -a 10.10.11.99
```

Access shares:

```
smbclient //10.10.11.99/public -N
```

Look for:

- Credential files
- Shell scripts

- Usernames
-

1.5 NFS (Port 2049)

```
showmount -e 10.10.11.99  
mount -t nfs 10.10.11.99:/home /mnt/nfs
```

Look for:

- SSH keys
 - Bash history
 - Cron jobs
-

1.6 RPC Services (Port 111)

```
rpcinfo -p 10.10.11.99  
nmap -sV --script=rpcinfo 10.10.11.99
```

If vulnerable versions found, check for exploits.

1.7 Misc Services

SNMP (161):

```
snmpwalk -v1 -c public 10.10.11.99
```

MySQL/PostgreSQL (3306, 5432):

```
nmap -p 3306 --script=mysql-info,mysql-users,mysql-databases 10.10.11.99
```

Try:

```
mysql -h 10.10.11.99 -u root -p
```

2. User Enumeration

SSH User Brute:

```
hydra -L users.txt -P rockyou.txt ssh://10.10.11.99
```

HTTP User Brute (form-based login):

```
wfuzz -c -w usernames.txt -w passwords.txt -d "user=FUZZ&pass=FUZZ" http://target/login
```

Search in found files:

- `cat *.txt | grep -i user`
- Look for usernames in share file paths: `/home/alice/` , `/var/www/bob/`

3. Password Enumeration

Look in exposed files:

- `.env`
- `.git/config`
- `config.php`
- `.bash_history`

Common patterns:

- Base64 strings (`echo "dXNlcjpwYXNz" | base64 -d`)
- Hardcoded creds in scripts

4. Automation & Toolkits

LinEnum (Post-shell) — useful to prep:

```
wget https://raw.githubusercontent.com/rebootuser/LinEnum/master/LinEnum.sh
```

AutoRecon (External & post-exploitation):

```
autorecon 10.10.11.99
```

PEASS-ng (Post-exploitation) — prep for upload:

```
wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
```

5. Manual Tips While Exploring

- Always check `robots.txt` , backup folders
- Enumerate subdomains (`ffuf` , `dnsrecon`)
- Check web extensions: `.php.bak` , `.swp` , `.old`

Summary Table

Port	Service	Tools to Use	Key Actions
22	SSH	<code>hydra</code> , <code>nmap</code> , <code>ssh</code>	Brute-force, weak creds
21	FTP	<code>ftp</code> , <code>nmap</code> , <code>hydra</code>	Try anonymous, download files
80/443	HTTP/HTTPS	<code>whatweb</code> , <code>nikto</code> , <code>ffuf</code>	Discover paths, test login, LFI
139/445	SMB	<code>smbclient</code> , <code>enum4linux</code> , <code>smbmap</code>	List shares, read files, find creds
2049	NFS	<code>showmount</code> , <code>mount</code>	Mount exports, look for home dirs
389	LDAP	<code>ldapsearch</code>	Query for users/groups
3306	MySQL	<code>nmap</code> , <code>mysql</code>	Check for open DBs
111	RPC	<code>rpcinfo</code> , <code>nmap</code>	Identify RPC services
5985	WinRM	<code>evil-winrm</code> (Windows)	Shell (Windows only)